

Time Series Final Project

Dataset and preprocessing The dataset comprises 2,106 daily observations for Victoria, Australia, with daily electricity demand as the target and min/max temperature, solar exposure, rainfall, retail price, school day, and holiday as predictors. After recoding the school day and holiday flags to binary, we split the series into a train set and a 98-day test set.

Classical decomposition Applying classical decomposition yields observed, trend, seasonal, and random components. The trend oscillates between roughly 100-140 GWh in annual waves with no long-term drift, and the seasonal panel shows a consistent weekly swing of about 15 GWh. The random component centres on zero within ± 10 GWh, with recurring annual excursions to -20/+30 GWh consistent with extreme-temperature days.

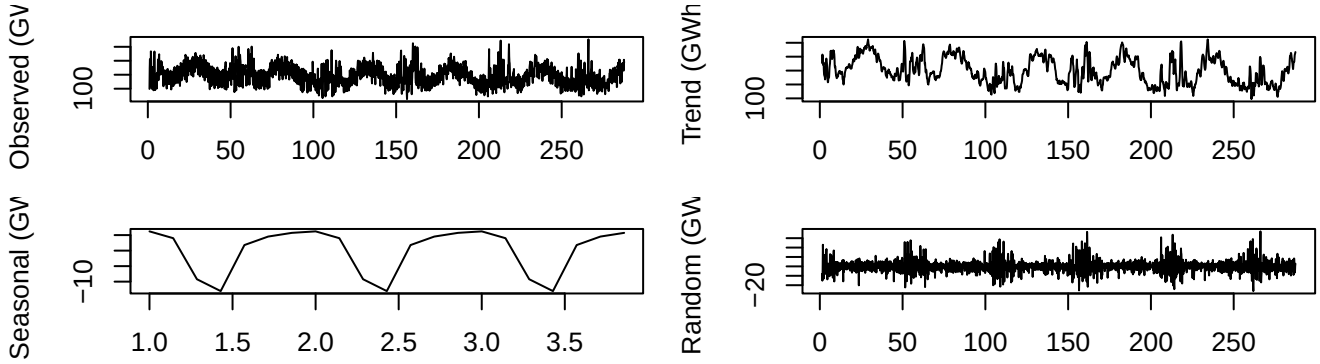


Figure 1: Classical decomposition of daily demand (in GWh).

Stationarity tests The table reports three complementary tests: ADF (H_0 :non-stationary), KPSS (H_0 :stationary), and Ljung-Box (H_0 :no autocorrelation). ADF rejects non-stationarity, while KPSS and Ljung-Box both reject their nulls, indicating residual non-stationarity from a seasonal component.

test	null	p_value	verdict
ADF	non-stationary	0.01	stationary
KPSS	stationary	0.01	non-stationary
Ljung-Box	no autocorrelation	0.00	non-stationary

ACF and PACF The ACF declines slowly from ≈ 0.85 at lag 1 with spikes at lags 7, 14, 21, 28, confirming the non-stationarity and weekly cycle. The PACF spikes at lag 1 (≈ 0.65), suggesting an AR(1) component and a weekly seasonal AR term. Both functions tail off rather than cut off, pointing to an ARMA model.

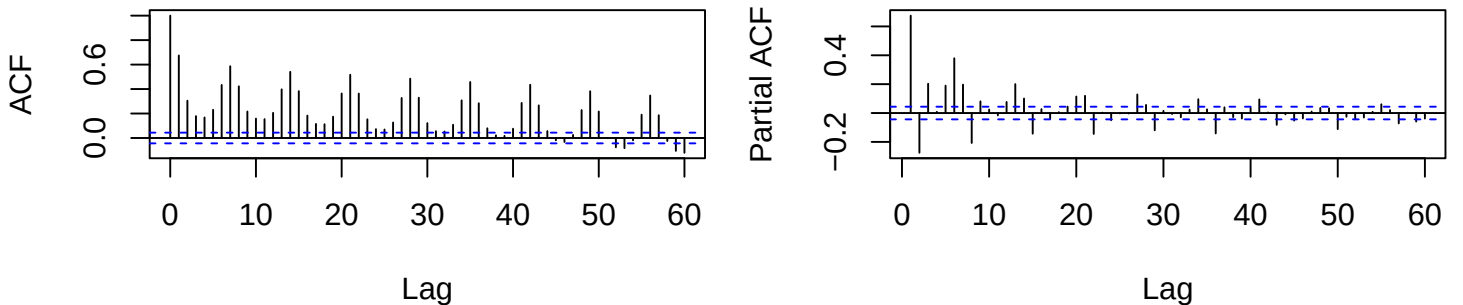


Figure 2: ACF (left) and PACF (right) of daily training demand, lags 0 to 60.

Demand vs temperature Plotting daily demand against minimum and maximum temperature. The maximum-temperature cloud reaches the hot extremes where demand responds most strongly, making it the stronger predictor and motivating a quadratic term.

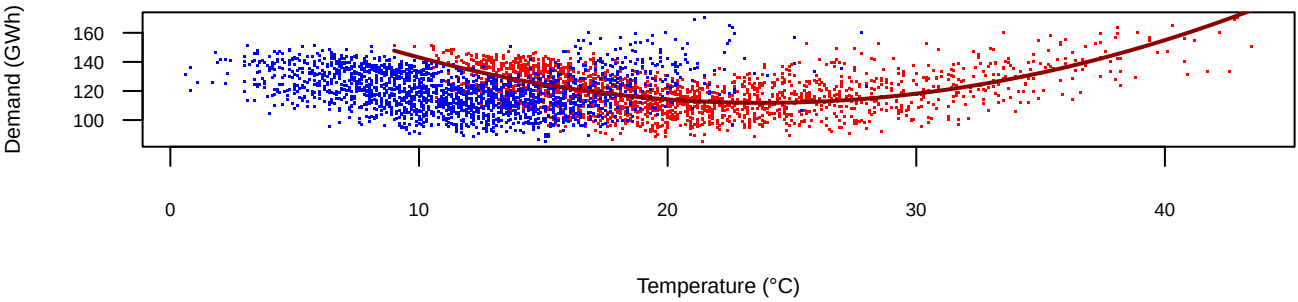


Figure 3: Daily demand against min/max temperature, with quadratic fit on max temperature.

Boxplots by school day and holiday Comparing demand distributions across calendar categories, school days show only marginal uplift with overlapping IQRs while holiday demand sits around 20 GWh below non-holiday with a tighter spread, marking holiday the stronger predictor.

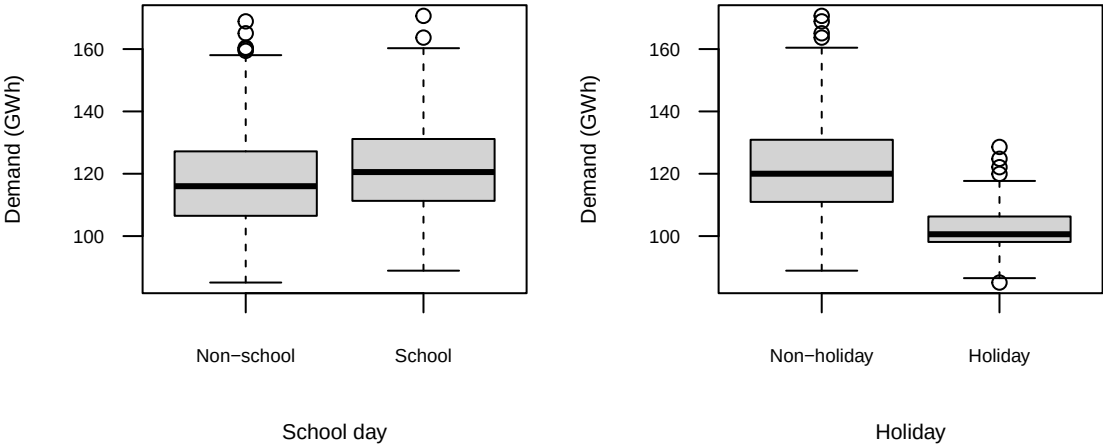


Figure 4: Distribution of daily demand (in GWh) by school day status (left) and holiday status (right).

Model selection and forecast We evaluate four candidates: seasonal naive, Holt-Winters, regression with ARIMA errors (temp, square, holiday, school), and dynamic harmonic regression with the same predictors plus Fourier terms for the weekly cycle.

Model selection DHR+xreg achieves the lowest AICc and BIC, confirming that ARIMA-based modelling outperforms H-W and that Fourier terms capture the weekly cycle more efficiently than a seasonal ARIMA component.

model	AICc	BIC
Holt-Winters	51356.0	51423.1
ARIMA + xreg	40280.2	40336.1
DHR + xreg	39787.8	39877.2

Residual diagnostics Residual diagnostics confirm the model has captured the structure: Ljung-Box does not reject white noise, the residual ACF lies within confidence bands at virtually every lag, the time plot is centred on zero, and the histogram is approximately symmetric.

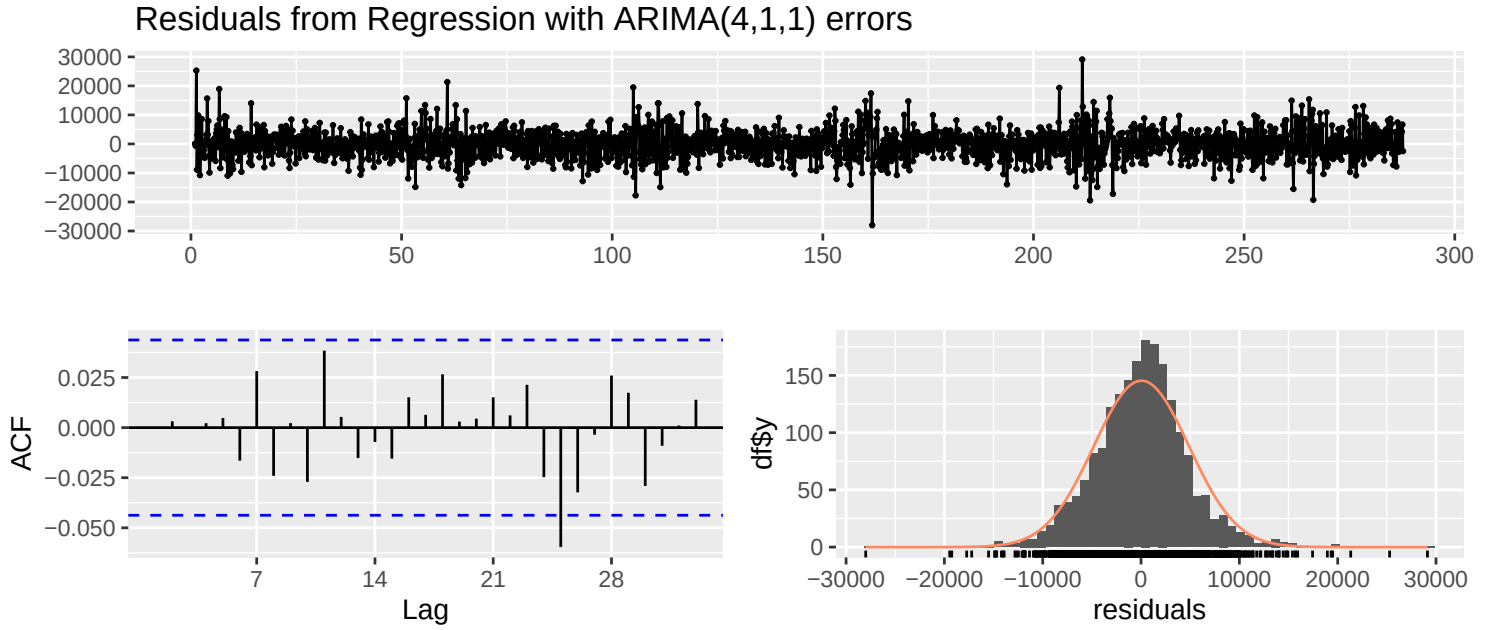


Figure 5: Residual time plot, ACF, histogram and Ljung-Box test for chosen model

Forecast vs actual demand

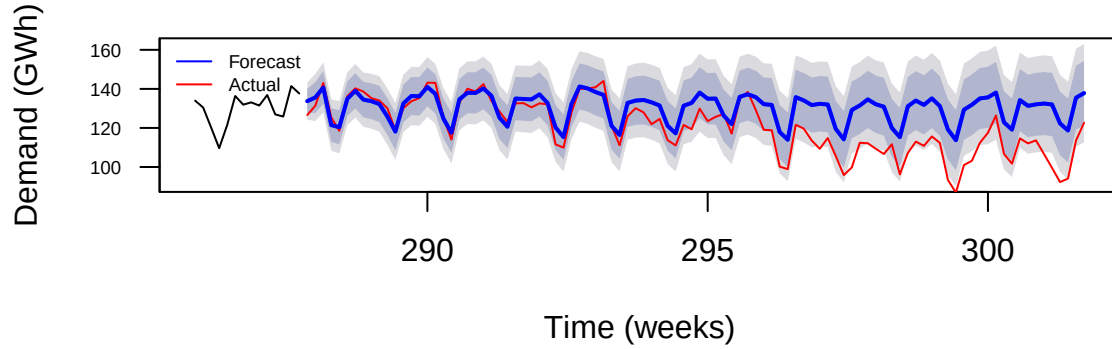


Figure 6: Forecast (blue) with 80% and 95% prediction intervals against actual test-set demand (red).

DHR+xreg achieves approximately 22% lower RMSE and 26% lower MAE/MAPE than the seasonal naive baseline, tracking the broad pattern of test-period demand with a few high-demand spikes exceeding the prediction intervals.

model	RMSE	MAE	MAPE
Seasonal naive	18134	14603	13.23
DHR + xreg	14117	10836	9.87

Strengths and Limitations DHR + xreg outperforms both the seasonal naive and Holt-Winters baselines across all accuracy metrics, and its residuals satisfy the white-noise assumption, lending credibility to the prediction intervals. A key limitation is that the test period coincides with the COVID-19 lockdown, introducing atypical demand patterns absent from the training data and not explicitly modelled. The forecast is also conditional on observed test-period weather, and at the 98-day horizon the prediction intervals widen substantially.

Appendix — R code

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knitr::opts_chunk$set(echo = TRUE, message = FALSE, warning = FALSE, fig.width = 7, fig.height =
library(forecast)
library(fpp2)
library(tseries)

# Read csv , and parse date column
dataset <- read.csv("data/elecDaily.csv")
dataset$date <- as.Date(dataset$date)

# Convert Y/N flags to binary
dataset$school_day <- ifelse(dataset$school_day == "Y", 1, 0)
dataset$holiday <- ifelse(dataset$holiday == "Y", 1, 0)

# Split: training ends 2020-06-30, test covers the final 98 days
train <- dataset[dataset$date <= as.Date("2020-06-30"), ]
test <- dataset[dataset$date >= as.Date("2020-07-01"), ]

# Convert demand to GWh (thousands of MWh) for cleaner y-axis labels
demand_train_weekly <- ts(train$demand / 1000, frequency = 7)

# Classical additive decomposition (weekly seasonality)
demand_decomp <- decompose(demand_train_weekly)

# Plot the four-panel decomposition
par(mar = c(3, 4, 1, 1), mfrow = c(2, 2))
plot(demand_decomp$x, ylab = "Observed (GWh)", xlab = "")
plot(demand_decomp$trend, ylab = "Trend (GWh)", xlab = "")
plot(window(demand_decomp$seasonal, end = c(3, 7)), ylab = "Seasonal (GWh)", xlab = "Week")
plot(demand_decomp$random, ylab = "Random (GWh)", xlab = "")

# Apply ADF, KPSS and Ljung-Box tests to the training demand series
adf_result <- adf.test(train$demand)
kpss_result <- kpss.test(train$demand)
box_result <- Box.test(train$demand, lag = 14, type = "Ljung-Box")

# Summarise p-values and the implied verdict
knitr::kable(data.frame(
  test = c("ADF", "KPSS", "Ljung-Box"),
  null = c("non-stationary", "stationary", "no autocorrelation"),
  p_value = round(c(adf_result$p.value, kpss_result$p.value, box_result$p.value), 4),
  verdict = c(
    ifelse(adf_result$p.value < 0.05, "stationary", "non-stationary"),
    ifelse(kpss_result$p.value > 0.05, "stationary", "non-stationary"),
    ifelse(box_result$p.value > 0.05, "stationary", "non-stationary")
  )
))

# Side-by-side ACF and PACF for the training demand series
par(mar = c(4, 4, 1, 1), mfrow = c(1, 2))
acf(train$demand, lag.max = 60, main="")
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pacf(train$demand, lag.max = 60, main="")

# Plot demand vs max temperature in red
par(mar = c(4, 4, 1, 1))
plot(train$max_temperature, train$demand / 1000, pch = ".", col = "red",
      xlab = "Temperature (\u00b0C)", ylab = "Demand (GWh)",
      xlim = range(c(train$min_temperature, train$max_temperature)),
      cex.axis = 0.6, cex.lab = 0.7, las = 1)

points(train$min_temperature, train$demand / 1000, pch = ".", col = "blue")

quad_fit <- lm(I(demand/1000) ~ max_temperature + I(max_temperature^2), data = train)
temp_seq <- seq(min(train$max_temperature), max(train$max_temperature), length.out = 100)
lines(temp_seq,
      predict(quad_fit, newdata = data.frame(max_temperature = temp_seq)),
      col = "darkred", lwd = 2)

# Side-by-side boxplots for the two binary calendar flags
par(mar = c(4, 4, 1, 1), mfrow = c(1, 2))

boxplot(demand / 1000 ~ school_day, data = train,
        names = c("Non-school", "School"),
        xlab = "School day", ylab = "Demand (GWh)",
        cex.axis = 0.6, cex.lab = 0.7, las = 1)

boxplot(demand / 1000 ~ holiday, data = train,
        names = c("Non-holiday", "Holiday"),
        xlab = "Holiday", ylab = "Demand (GWh)",
        cex.axis = 0.6, cex.lab = 0.7, las = 1)

# Wrap demand for modelling (weekly frequency for seasonal handling)
demand_ts <- ts(train$demand, frequency = 7)

# Model A: seasonal naive - forecast = same day last week
fit_snaive <- snaive(demand_ts, h = 98)

# Model B: Holt-Winters exponential smoothing (level + trend + weekly seasonal)
fit_hw <- hw(demand_ts, h = 98)

# Predictors chosen from data analysis: U-shape => temp + temp^2; calendar effects
xreg_train <- cbind(
  max_temp      = train$max_temperature,
  max_temp_sq   = train$max_temperature^2,
  holiday       = train$holiday,
  school_day    = train$school_day
)

# Model C: regression with ARIMA errors using the EDA-driven predictors
fit_xreg <- auto.arima(demand_ts, xreg = xreg_train)

# Model D: dynamic harmonic regression with K=3 Fourier terms for the weekly

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# cycle, plus the same predictors as Model C; seasonal=FALSE turns off the
# seasonal ARIMA term since Fourier handles the seasonality
fit_dhr <- auto.arima(
  demand_ts,
  xreg = cbind(fourier(demand_ts, K = 3), xreg_train),
  seasonal = FALSE
)

# Compare the three likelihood-fitted models (SNAIVE has no AICc/BIC
# because it has no parameters; included via test-set accuracy below).
knitr::kable(data.frame(
  model = c("Holt-Winters", "ARIMA + xreg", "DHR + xreg"),
  AICc = round(c(fit_hw$model$aicc, fit_xreg$aicc, fit_dhr$aicc), 1),
  BIC = round(c(fit_hw$model$bic, fit_xreg$bic, fit_dhr$bic), 1)
))

checkresiduals(fit_dhr)

# Build test-set predictors with the same names as training xreg
xreg_test <- cbind(
  max_temp = test$max_temperature,
  max_temp_sq = test$max_temperature^2,
  holiday = test$holiday,
  school_day = test$school_day
)

# Generate the 98-day forecast for DHR
# Fourier terms must be projected forward with h=98 and combined with test xreg
fc <- forecast(fit_dhr,
  xreg = cbind(fourier(demand_ts, K = 3, h = 98), xreg_test),
  h = 98)

# Wrap test demand as ts continuing the training timeline
test_ts <- ts(test$demand,
  start = end(demand_ts) + c(0, 1),
  frequency = 7)

# Plot training tail + forecast + actuals
par(mar = c(4, 4, 1, 1))
plot(fc, main="", xlab = "Time (weeks)", ylab = "Demand (GWh)",
  cex.axis = 0.6, cex.lab = 0.7, las = 1, yaxt = "n",
  include = 14)
ticks <- pretty(c(fc$x, fc$mean, test_ts))
axis(2, at = ticks, labels = ticks / 1000, cex.axis = 0.6, las = 1)
lines(test_ts, col = "red")
lines(fc$mean, col = "blue", lwd = 2)
legend("topleft", legend = c("Forecast", "Actual"),
  col = c("blue", "red"), lty = 1, bty = "n", cex = 0.6)

# Evaluate the chosen model and the seasonal naive baseline on the test set
acc_dhr <- accuracy(fc, test$demand)

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acc_snaive <- accuracy(fit_snaive, test$demand)

knitr::kable(data.frame(
  model = c("Seasonal naive", "DHR + xreg"),
  RMSE = round(c(acc_snaive["Test set", "RMSE"], acc_dhr["Test set", "RMSE"]), 0),
  MAE = round(c(acc_snaive["Test set", "MAE"], acc_dhr["Test set", "MAE"]), 0),
  MAPE = round(c(acc_snaive["Test set", "MAPE"], acc_dhr["Test set", "MAPE"]), 2)
))

```